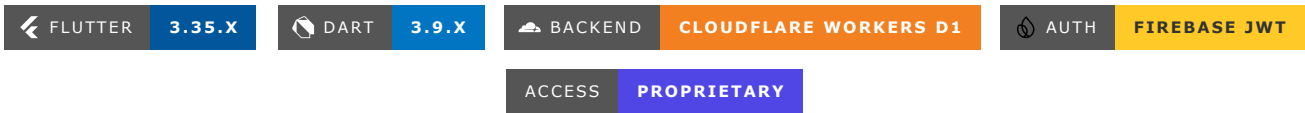




# AcePharm Mobile Companion App — Technical Implementation Plan

Cross-Platform iOS & Android Companion Client (Flutter / Dart 3.x)



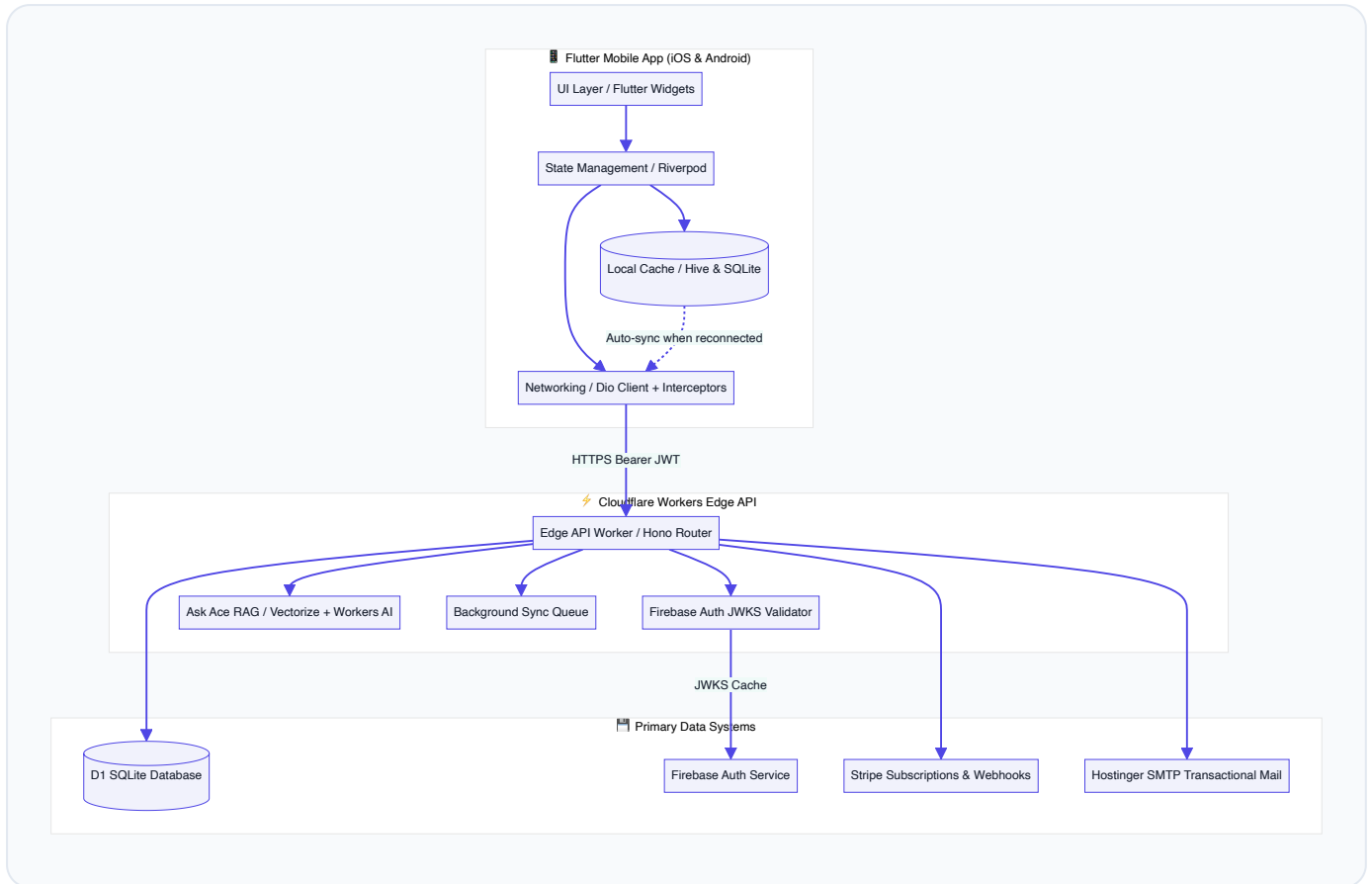
## Executive Summary & Grounding

This document outlines the end-to-end technical blueprint, architecture, design system translation, and release roadmap for the **AcePharm Mobile Companion App** ( [mobile-app/](#) ).

The mobile client is engineered directly from the requirements in [Brief\\_AcePharmMobileApp\\_Numan.docx](#), grounded in the **Developer Brief v2.1** and **Website Copy v2.0**, and communicates seamlessly with the unified [web-app](#) backend infrastructure.

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## System Architecture



## Design System & Token Bridge (Section 3.3 Translation)

All visual styling is translated directly from the web design tokens ( `tokens.css` ) into a strongly typed `AceTheme` in Flutter, ensuring pixel-perfect fidelity without approximations.

### Color Tokens

| Token Name         | Hex Code | Flutter Color Definition       | Application / Purpose                                              |
|--------------------|----------|--------------------------------|--------------------------------------------------------------------|
| <b>Ace Indigo</b>  | #4F46E5  | <code>Color(0xFF4F46E5)</code> | Primary brand accent, selected active states, focus rings          |
| <b>Deep Indigo</b> | #3730A3  | <code>Color(0xFF3730A3)</code> | Pressed/Active interactive button states, high-contrast text       |
| <b>Indigo Wash</b> | #F1F2FC  | <code>Color(0xFFF1F2FC)</code> | Pill backgrounds, subtle card tints, active navigation backgrounds |
| <b>Ink</b>         | #111827  | <code>Color(0xFF111827)</code> | Primary typography, question stems, lead-ins, headings             |

| Token Name  | Hex Code | Flutter Color Definition | Application / Purpose                                    |
|-------------|----------|--------------------------|----------------------------------------------------------|
| Slate       | #64748B  | Color(0xFF64748B)        | Secondary descriptions, subheadings, unselected options  |
| Slate Light | #94A3B8  | Color(0xFF94A3B8)        | Helper captions, metadata dates, question IDs            |
| Teal        | #0F766E  | Color(0xFF0F766E)        | Correct answer cues, positive calibration, active badges |
| Teal Light  | #EFAF8   | Color(0xFFEFAF8)         | Correct answer background container, pill tint           |
| Danger Rose | #B91C1C  | Color(0xFFB91C1C)        | Incorrect answer cues, danger warnings, account deletion |
| Canvas      | #F8F8F8  | Color(0xFFF8F8F8)        | Primary app screen background                            |
| Surface     | FFFFFF   | Color(0xFFFFFFFF)        | Card containers, modal sheets, question option panels    |
| Border      | #E2E8F0  | Color(0xFFE2E8F0)        | Subtle hairline card borders, dividers, option outlines  |

Typography & Spacing Scale

Note

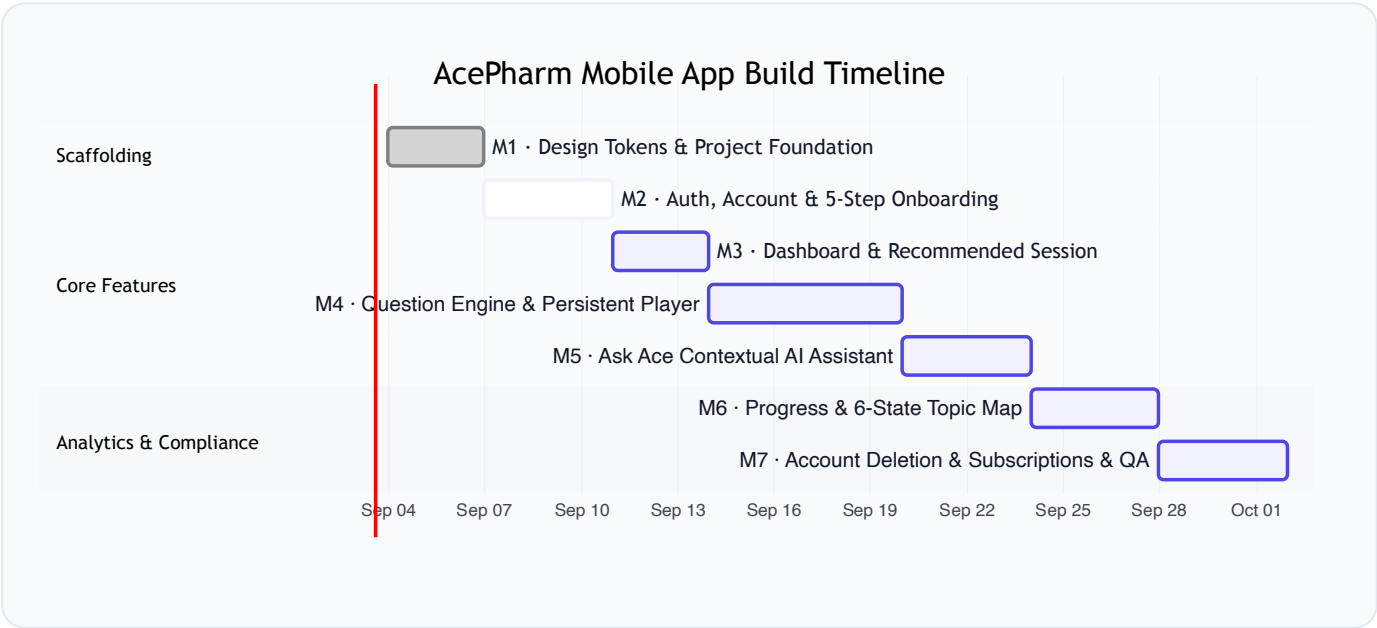
- **Primary Font Family:** Geist Sans (via google\_fonts ) for all interface copy, prompts, and clinical vignettes.
- **Monospace Font Family:** Geist Mono for question IDs ( ACP-CV-0012 ), calculations, and mathematical dosages.
- **Touch Targets:** Minimum 44x44 pt enforced across all option rows, buttons, and icon tap targets.
- **Spacing:** Strict 8-point base scale ( 4px , 8px , 12px , 16px , 24px , 32px , 48px ).

⚡ Product Invariants & Non-Negotiables

These rules are strictly enforced across the mobile companion app to preserve clinical educational integrity:

| CRITICAL PRODUCT RULES             |                                                                                                               |
|------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 1. Zero Pre-Submission Color       | Options remain neutral upon selection. Absolutely no green/red visual hints before clicking Submit.           |
| 2. Mandatory Confidence Catch      | Confidence rating (High / Med / Low) must be captured prior to submission (unless disabled in settings).      |
| 3. Multi-Cue A11y Feedback         | Post-submission feedback must include an icon + text label + color (e.g. "✓ Correct" / "✗ Incorrect").        |
| 4. Individual Distractor Rationale | Every incorrect choice receives its own dedicated rationale attributed to "AcePharm Clinical Editorial".      |
| 5. Five Discrete Metrics           | 1st-Attempt, Practice, Repeat, Calibration & Coverage are kept visibly distinct. Never merged into one score. |
| 6. Offline Answer Resilience       | No answer is ever lost on network drop. Answers queue locally and sync automatically when internet restores.  |

## Milestone Roadmap & Implementation Steps



### ◆ Milestone 1: Project Scaffolding & Design System Bridge

- Initialize Flutter project under `mobile-app/` ( `com.acepharm.app` ).
- Implement `AceTheme` , `AceColors` , `AceTypography` , and `AceSpacing` .
- Build reusable UI primitives: `AceButton` , `AceCard` , `AceBadge` , `AceInput` , `AceModalSheet` .

### ◆ Milestone 2: Auth, Account & 5-Step Onboarding

- Firebase Auth integration with Google JWKS-backed token validation.
- Login, Sign-up, Password Reset (with real-time password entropy meter).

- **5-Step Onboarding Flow:**

1. Training stage selection (MPHarm Year 2–4, Foundation, Oriel, IP).
2. Primary revision target.
3. Exam date countdown ticker.
4. Daily question goal (default: 20).
5. University / Institution affiliation.

- ◆ **Milestone 3: Dashboard & Recommended Next Session**

- **Real-Time Revision Ring:** Circular daily target progress tracker (e.g., *14/20 completed*).
- **Weekly Ace Clinical Insight:** Dynamic clinical takeaways delivered via Cloudflare D1.
- **Explainable Recommendation Engine:**

*"Recommended because your first-attempt accuracy in Cardiovascular Therapeutics is 44%."*  
(Never an unexplained session recommendation).

- ◆ **Milestone 4: Question Practice Engine (The Core Feature)**

- **Session Builder:**
  - Mode Selection: **Learn Mode** (untimed with immediate multi-stage rationales) vs **Timed Exam Mode** (Pearson VUE simulated countdown).
  - Curriculum Category filter across all 19 GPhC domains.
- **Question Screen UX:**
  - Pre-submission confidence selector.
  - Active recall **"Cover Options"** toggle.
  - Subtopic personal clinical notes & question bookmarking.
  - Isolated first-attempt telemetry vs subsequent practice attempts.

- ◆ **Milestone 5: Contextual "Ask Ace" AI Clinical Tutor**

- Integrated *"Ask Ace about this question"* sheet beneath explanation cards.
- Streamed responses from `/api/v1/ace/chat` via Cloudflare Workers AI.
- Explicit guideline citations (BNF & NICE NG guidelines).
- Graceful refusal when clinical queries fall outside the reviewed curriculum.

- ◆ **Milestone 6: Progress & Diagnostic Calibration**

- **5 Separate Analytics Pillars:**
  1. *First-Attempt Accuracy* (prominently featured).
  2. *Practice Accuracy* (inclusive of second/third attempts).

3. *Repeat Accuracy*.

4. *Confidence Calibration Grid* (Overconfident vs Calibrated vs Underconfident).

5. *Curriculum Coverage Percentage*.

- **19-Topic Mastery Grid** with 6 status labels:

Not started → Firstpass → Needs attention → Developing → Secure → Due for review

## ◆ Milestone 7: Account Management, Account Deletion & Store Release

- **Subscription Status Screen:** Displays active plan tier, renewal date, and deep link to web customer portal.
- **In-App Account Deletion Flow (Apple Guideline 5.1.1(v) & GDPR Compliance):**
  - Accessible via *Settings* → *Account* → *Close / Delete Account*.
  - 2-step confirmation modal detailing permanent data loss.
  - Re-authentication security check before issuing `DELETE /api/v1/user/account`.
  - Local cache wipeout (Hive & secure storage) and graceful exit to the onboarding screen.
- **Distribution Setup:** Fastlane build automation for Google Play Internal Testing and Apple TestFlight.

## 📁 Directory Structure ( `mobile-app/` )

```
mobile-app/
├── android/           # Native Android wrapper & Gradle configs
├── ios/              # Native iOS Xcode workspace & Podfiles
├── assets/
│   ├── fonts/        # Geist-Regular, Geist-Bold, GeistMono-Medium
│   └── icons/         # Brand SVG marks and status glyphs
├── lib/
│   ├── core/
│   │   ├── api/       # Dio HTTP client, JWT interceptor, retry logic
│   │   ├── database/  # Hive local database & offline answer queue
│   │   ├── theme/     # AceTheme, AceColors, AceTypography
│   │   └── utils/     # Formatters, connectivity watchers
│   ├── features/
│   │   ├── auth/      # Login, register, password reset, action handler
│   │   ├── onboarding/ # 5-step onboarding carousel
│   │   ├── dashboard/ # Daily goal ring, streak counter, recommendations
│   │   ├── practice/  # Question player, rationale sheet, session summary
│   │   ├── ace/       # Ask Ace AI streaming chat bottom sheet
│   │   ├── progress/  # 5-metric dashboard, 19-category status map
│   │   └── settings/  # Account, deletion modal, preferences, subscription
│   └── main.dart      # App bootstrap & Riverpod ProviderScope
└── pubspec.yaml      # Flutter dependencies & assets declarations
```

## Comprehensive Verification & QA Matrix

| MANUAL QA VERIFICATION PLAN |                         |                                                                                 |        |
|-----------------------------|-------------------------|---------------------------------------------------------------------------------|--------|
| Ref                         | Feature / Flow          | Expected Behaviour                                                              | Status |
| QA-1                        | Pre-Submit Confidence   | Submit button is disabled until a confidence rating is set.                     | [ ]    |
| QA-2                        | Zero Pre-Submit Color   | Selected option remains neutral border. No green/red cues.                      | [ ]    |
| QA-3                        | Distractor Rationales   | Every wrong answer option reveals an independent rationale.                     | [ ]    |
| QA-4                        | Offline Mode Resilience | In Airplane mode, submitted answers queue without error.                        | [ ]    |
| QA-5                        | 5 Progress Metrics      | First-attempt accuracy is visibly distinguished.                                | [ ]    |
| QA-6                        | Account Deletion        | Account deletion wipes D1 data, purges cache & logs out.                        | [ ]    |
| QA-7                        | Ask Ace Citations       | Ask Ace provides BNF / NICE citations and refuses out-of-scope clinical advice. | [ ]    |

### Automated CLI Verification Commands

```
# 1. Analyze and check lint rules
flutter analyze

# 2. Check formatting
dart format --output=none --set-exit-if-changed .

# 3. Run all unit and widget tests
flutter test --coverage

# 4. Build release artifacts
flutter build apk --release      # Android APK
flutter build appbundle --release # Android Play Bundle
flutter build ipa --release      # iOS TestFlight Archive
```

**AcePharm UK Revision Platform** · Built by pharmacists for pharmacy students.

Documentation maintained under [docs/](#) in [acepharm-web-app](#) .